Observe. Pick. Place. Repeat.

Movable-Tray Table Clearing

YOLO tray segmentation around the existing bowl grasp-and-place pipeline.

01 Re-localize the tray

02 Reuse the grasp and motion

03 Verify after every placement

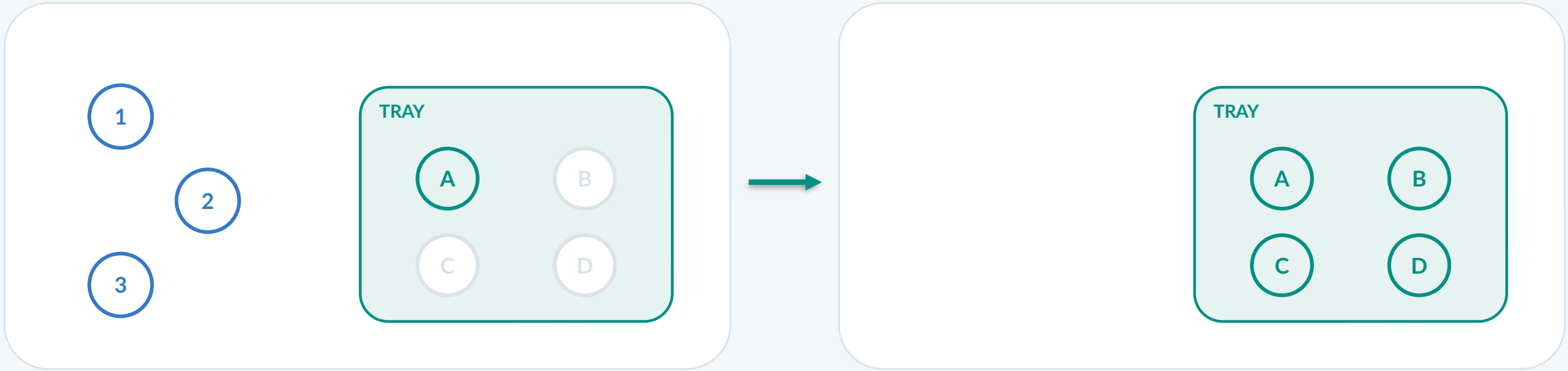


Historical project photograph • not the new tray demo

Three decisions make the task work

CURRENT SCENE

DESIRED SCENE



01

Where is the tray?

Update its position and orientation.

02

Which bowl is next?

Exclude bowls already inside.

03

Where can it go?

Choose an unoccupied interior slot.

One bowl per cycle. One fresh scene for the next decision.

01

Observe

Acquire a fresh RGB-D frame.

02

Measure

Locate the tray and all bowls.

03

Verify

Check the previous placement.

04

Choose

One source bowl + one free slot.

05

Pick & place

Run the existing motion sequence.

06

Return

Go back to the saved home pose.

First cycle: no previous placement to verify. Final cycle: observe again before declaring completion.

Add scene decisions around the existing pipeline

VISION

YOLO + RGB-D

Tray mask + original bowl clouds

→

GEOMETRY

Rim locator

Observed grasp candidates

→

DECISION

Existing runner

Filter bowls; choose tray slot

→

MOTION

Grasp executor

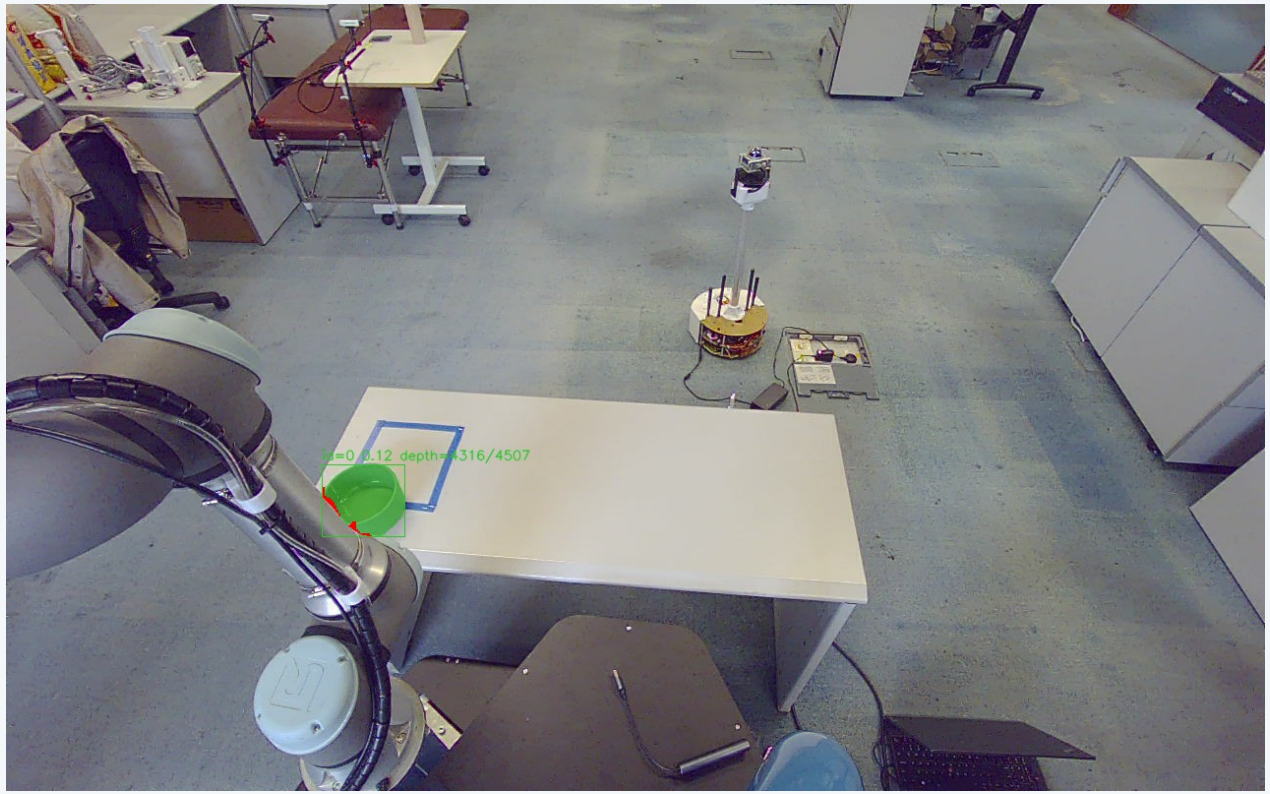
Plan, execute and return

Small additions

Tray localization • Bowl footprints • Occupancy memory • Placement confirmation

Shared acquisition stamp + task identity keep observations and grasp candidates aligned.

Turn visible bowls into physical footprints



HISTORICAL PROJECT IMAGE

Bowl footprint ≠ a single rim grasp point

RGB tells us what belongs to a bowl

YOLO provides the bowl mask.

Depth tells us where the surface is

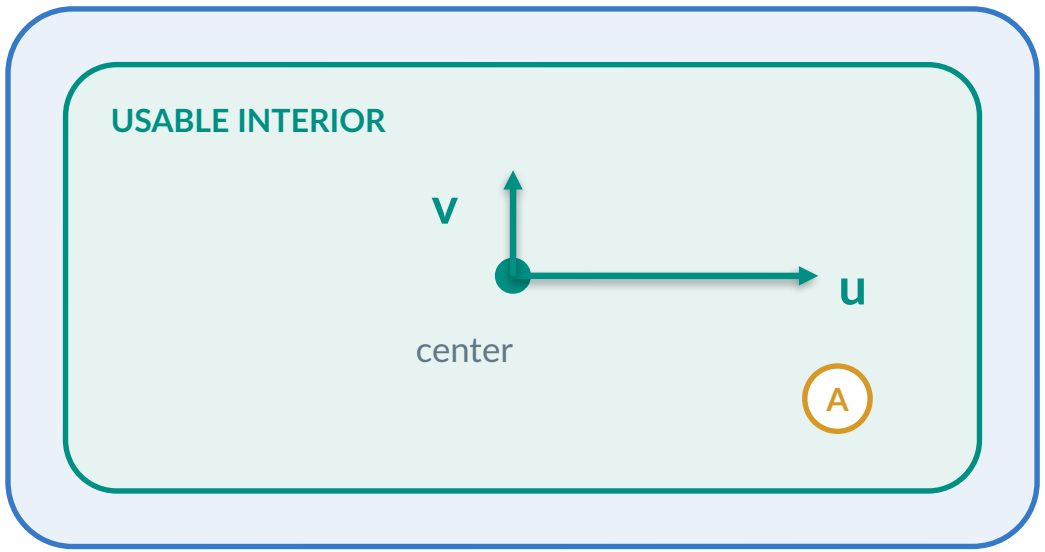
Use registered depth from the same frame.

Use a common coordinate frame

Transform to base; estimate center and radius.

Historical segmentation image • footprint estimation is the new task-level input

YOLO gives the tray a movable coordinate frame



Depth converts pixels to base XY.

Local slots follow the observed pose.

Segment the whole tray

Dedicated tray weights required; not included.

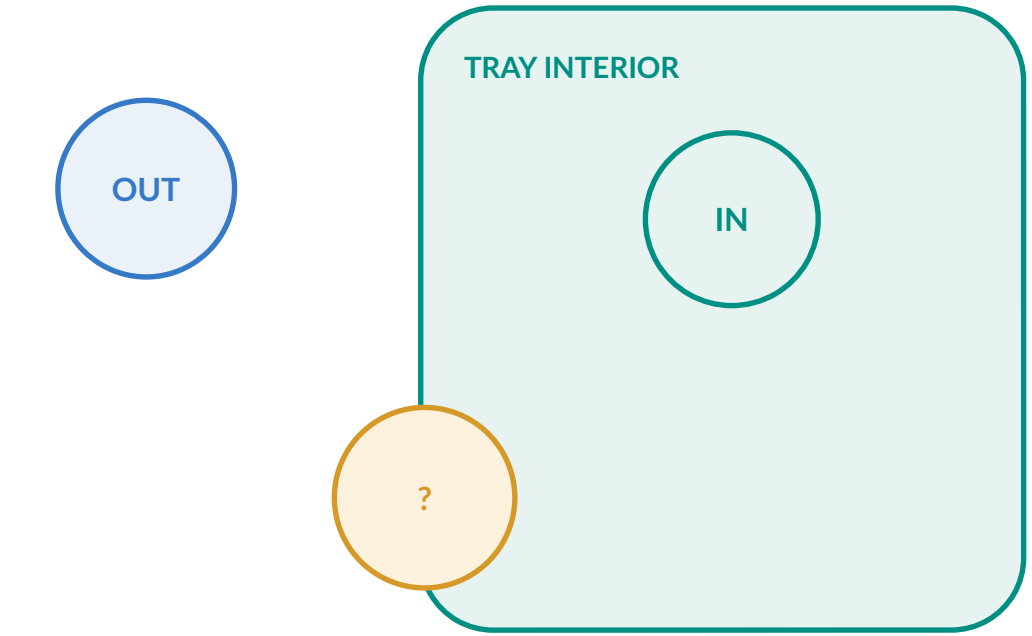
Measure center and long axis

Same-frame depth + acquisition-time TF.

Move the slots with the tray

$p_{slot} = c_{tray} + R(\theta) \cdot [u, v]$

A bowl is handled only when its footprint is inside



Inside → keep as occupancy

Do not submit this bowl for grasping.

Outside + in source region → candidate

Choose one using the existing ordering.

Uncertain → observe again

Boundary overlap or unreliable depth stays unresolved.

Select space for the actual bowl being carried



Used or uncertain slots are not immediately freed.

Check the tray edge

distance to edge ≥ bowl radius + margin

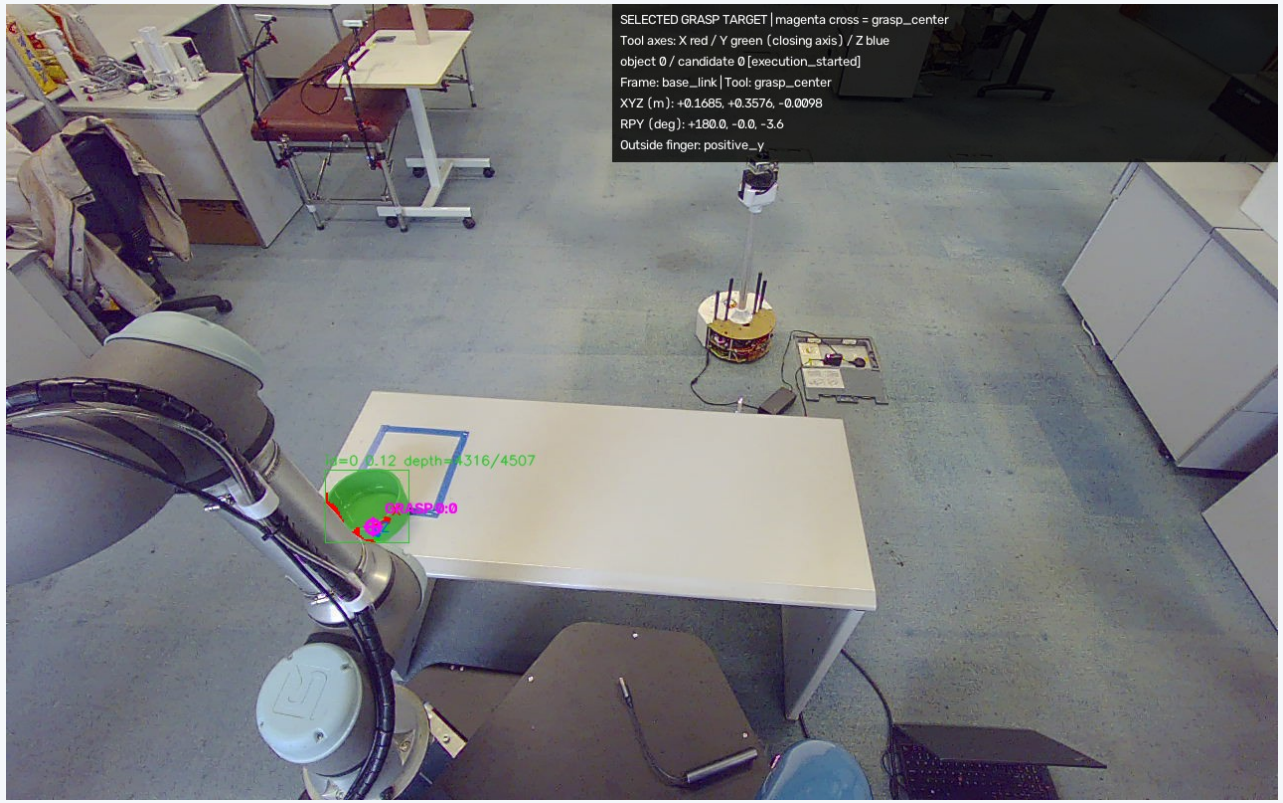
Check nearby bowls

center distance > both radii + clearance

Keep a small occupancy record

Reserve used slots; refresh observed occupancy.

Reuse the existing rim-grasp method



EXISTING GRASP PIPELINE

New task decisions do not redesign the grasp.

Use observed rim contacts

Fit reliable bowl geometry; retain measured contact points.

Keep finger direction consistent

Radial direction assigns the outside finger; tool Z points down.

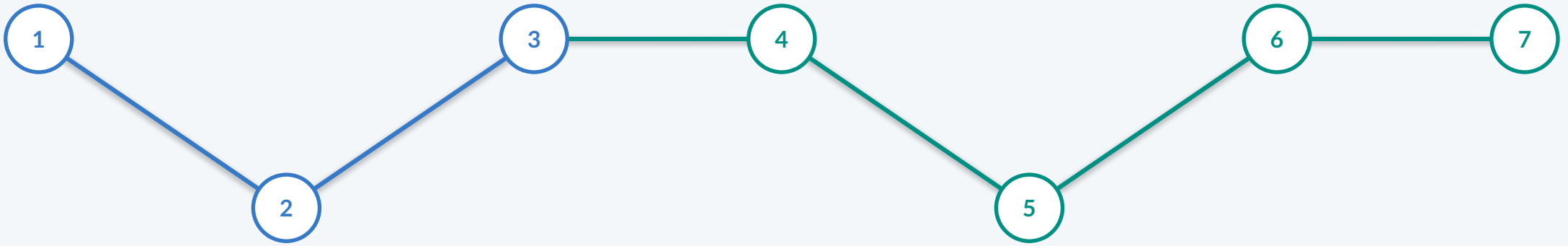
Preserve candidate fallback

Supported-wall fallback and pre-motion planning checks remain.

Historical grasp-target overlay • a planned/selected pose is not proof of a successful physical grasp

Keep the motion sequence; update the destination XY

SCHEMATIC MOTION ORDER • NOT A ROBOT TRAJECTORY

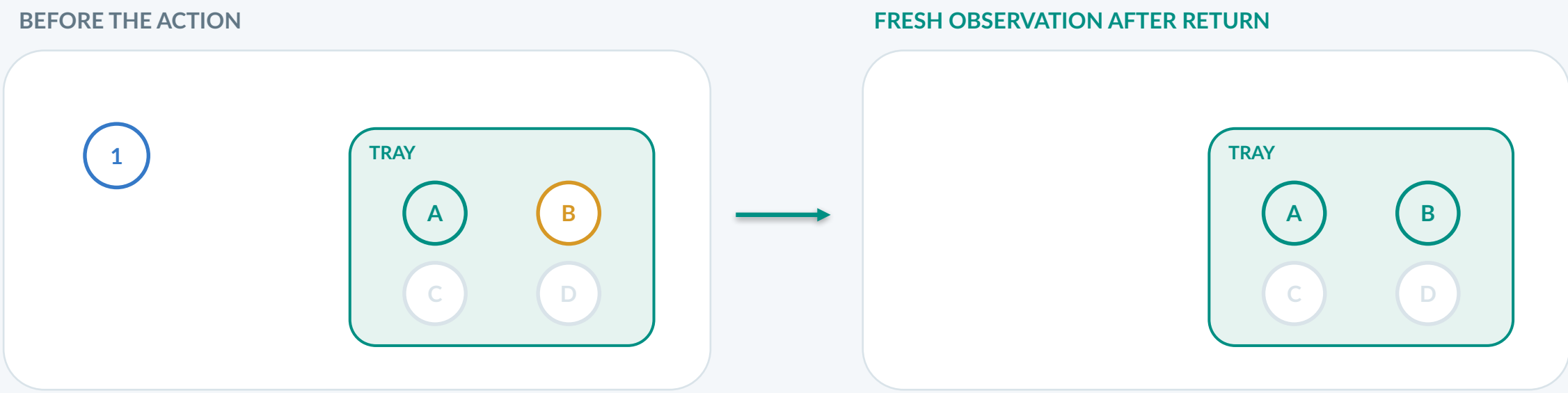


Approach	Descend to grasp	Lift	Transfer to pre-release	Descend and release	Retreat	Return home

tool release XY = selected slot XY + fixed offset

Same Z • orientation • descent • home

Confirm a changed scene before starting the next pick



1

Source bowl is gone

2

New occupancy at the chosen slot

3

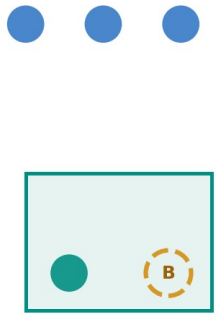
Previous tray occupancy remains

An action acknowledgment reserves the slot. A fresh observation confirms placement.

Three placements, five observations

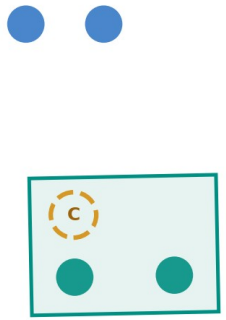
Blue: pending bowls • Teal: inside tray • Amber ring: selected destination

OBSERVATION 1



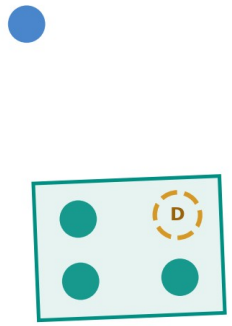
Pick → B

OBSERVATION 2



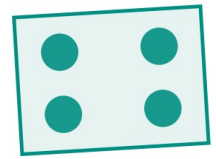
Confirm 1
Pick → C

OBSERVATION 3



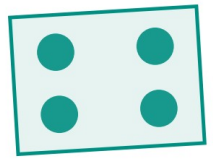
Confirm 2
Pick → D

OBSERVATION 4



Confirm 3
Check empty

OBSERVATION 5



Confirm empty
Complete

3 bowls confirmed / Slots B, C, D used / Existing slot A preserved

Four checks protect the YOLO tray workflow

01 Model contract

The existing bowl model has no tray class.

Require local tray segmentation weights and the correct class.

02 Full visibility

A tray detection does not prove the whole scene is visible.

Check contour, depth and the full tray / source view.

03 Axis continuity

A rectangular mask can reverse its reported axis.

Align equivalent axes; preserve occupied and reserved slots.

04 Actual bowl size

A fixed small-bowl assumption can select a tight slot.

Check this bowl against the edge and neighboring bowls.

Validation covers logic, ROS contracts and motion planning

888

automated tests passed

881 unit / regression

+ 7 isolated ROS control tests

3 ROS packages built

14 **ROS transport**

Fresh stamps, identities and retained messages.

32 **Launch contracts**

Correct topics, namespaces and runtime parameters.

2 **Interruption cleanup**

SIGTERM / SIGHUP leave no owned workers running.

16 **MoveIt integration**

Planning, candidate fallback and simulated return home.

From software validation to hardware trials

01

Prepare

Provide tray segmentation weights;

measure outer / usable tray sizes.

02

Preview

Calibrate the source and release;

check full visibility and the path.

03

Measure

Test one bowl first; record actual

placement error and repeatability.

Operating assumptions

One non-square rectangular tray • stationary during each action

No abrupt half-turns or tray swaps • no new 3D collision scene.

A simple loop with explicit evidence at every step

01 **Re-localize**

Use the tray pose from the current observation.

02 **Reserve space**

Check real footprints and remember occupied slots.

03 **Verify before continuing**

Confirm the last bowl before picking the next one.

The original grasp and placement motion remain the execution backbone.

Evidence, assets and reproducible entry points

DESIGN & IMPLEMENTATION

Minimal-change design
implement_new2.md

Operational guide
New_run/README.md

Tray localization and occupancy
src/yolo_vision/yolo_vision/tray_utils.py

Round decisions and placement mapping
tools/place_rounds.py

RECORDED EVIDENCE

Validation summary, test log and offline run
debug_output/clear_table_validation/yolo_20261010/

Historical photographs and overlays
Reused from the project's earlier presentation assets; all are labeled as historical in this deck.

Offline demo: ./run_clear_table.sh --offline-demo